KanalAtlas — Neighbourhood Accessibility and Scenario Assessment Report

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This report was generated automatically from the current on-screen state of the KanalAtlas prototype. It is an evidence summary for discussion, not a planning decision.

1. Study area

Seyhan Central Canal Corridor (Seyhan, Adana). The east–west corridor in Seyhan district mapped as an open irrigation canal in OpenStreetMap, and the surrounding residential area. Bounding box: S 36.999, W 35.268, N 37.022, E 35.33.

2. Background

Open irrigation canals run through residential areas of Adana. In its June 2026 decision on the “Yüzme Bilmeyen Çocuk Kalmasın” programme, Adana Metropolitan Council listed preventing drownings in unsafe places such as irrigation canals during the summer heat among the reasons for free swimming lessons. In September 2025 the Ministry of Agriculture and Forestry announced that the switch to closed, pressurised irrigation had begun and that open canals through the city centre will be decommissioned. KanalAtlas measures straight-line access to mapped public recreation facilities in this corridor and compares hypothetical improvements.

3. Analysis settings and existing facilities

- Category: Swimming facility
- Data scope: Core data (named/tagged records)
- Distance threshold: 1.00 km (straight line)
- Mapped facilities in the analysis: 5
- Sample map points: 72 (500 m grid)

Facility (nearest to the most points)	Status	Nearest for points	OSM source
Unnamed record	MAPPED	42	node/6833710832, way/739041056
Pınar Yüzme Havuzu	MAPPED	27	way/1272205437, way/1272205438
Yüzme Havuzu	MAPPED	3	way/1064356997

4. Sources and verification status

Geographic data: OpenStreetMap contributors (one-time Overpass API extract). Licence: ODbL 1.0 — © OpenStreetMap contributors. Extracted: 2026-10-22T20:38:43Z. None of the mapped facilities is marked 'VERIFIED': being in OSM does not mean they operate or are open to the public. Access-restricted records (access=private/customers) are excluded.

5. Hypothetical intervention

- Hypothetical swimming facility 1 (HYPOTHETICAL) — location 37.00555, 35.30858, near Namık Kemal. It becomes the nearest facility for 38 sample points; 13 of them newly fall within the threshold. Distance to the nearest existing facility of the same category: 2.32 km (unnamed).

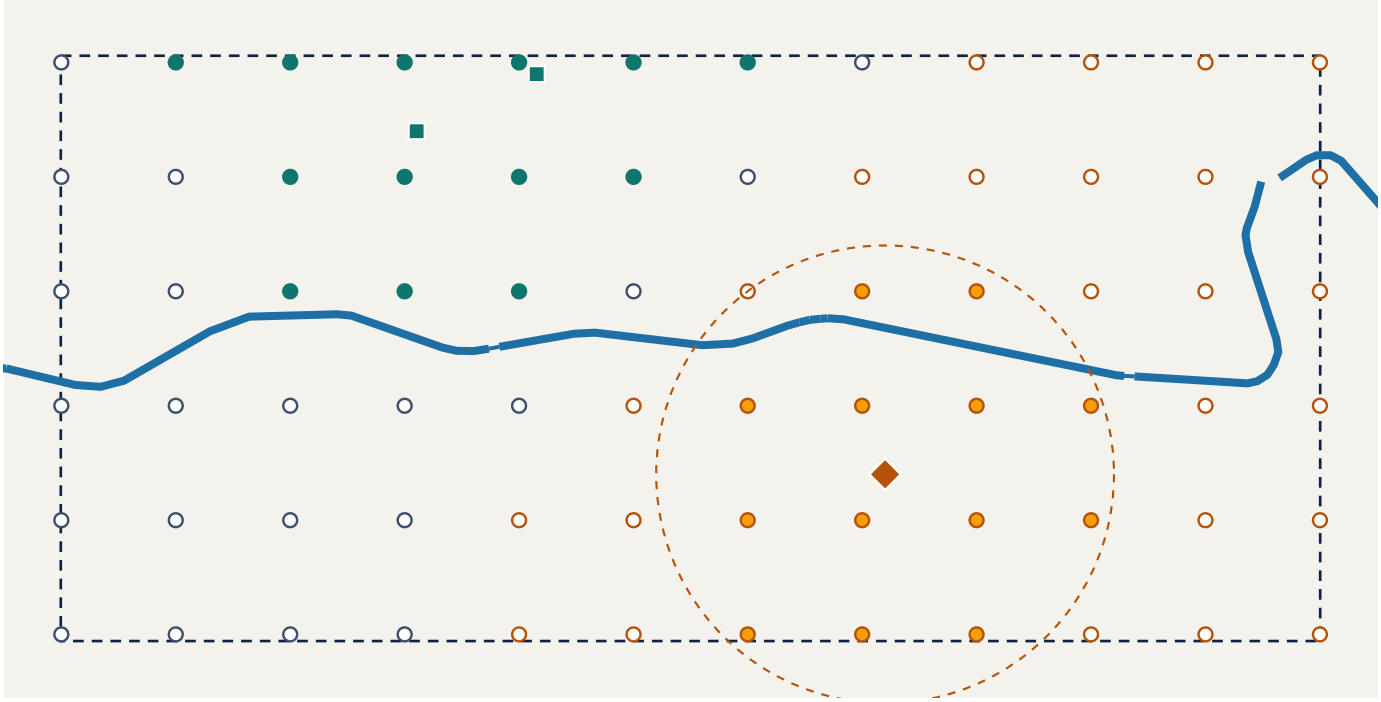
6. Before / after comparison

Measure (all sample points)	Baseline	Scenario
Sample points within 1.00 km	13 / 72	26 / 72
Mean nearest distance	1.99 km	1.28 km
Median nearest distance	1.97 km	1.23 km
Farthest sample point	4.21 km	2.69 km
Points with a defined distance	72	72

Canal belt (points ≤ 500 m from a canal)	Baseline	Scenario
Sample points within 1.00 km	3 / 27	9 / 27
Mean nearest distance	2.06 km	1.30 km
Median nearest distance	1.72 km	1.20 km

At baseline, 13 of 72 sample map points are within 1.00 km (straight line) of the nearest mapped swimming facility. In the hypothetical scenario this becomes 26 (+13). The nearest-facility distance gets shorter for 38 points; the largest reduction is 2.31 km. Mean distance: 1.99 km at baseline, 1.28 km in the scenario. These are straight-line distances, not walking times, and they do not represent population.

Schematic map (drawn from the app's coordinates)



Dashed frame: study area · blue line: mapped canal (thin: culvert/tunnel) · green square: existing facility in the analysis · circle: sample point (filled = within threshold, hollow = outside, amber = closest to a hypothetical facility) · amber diamond and dashed circle: hypothetical facility and threshold radius; thick amber line: hypothetical canal-line park.

7. Comparison of saved scenarios

All scenarios were recalculated with the same category, data scope and threshold at the time of the report.

Scenario	Location(s)	Within threshold	Mean	Points improved
Baseline	—	13 / 72	1.99 km	—
A	37.00555, 35.31759	27 / 72 (+14)	1.24 km	33
B	37.00555, 35.30858	26 / 72 (+13)	1.28 km	38

8. Priorities chosen by the user

The user did not enter priorities.

9. Method

1. The study area is a rectangle of about 5.5 km × 2.6 km around the Seyhan central corridor mapped as an open irrigation canal in OpenStreetMap.
2. A regular grid with 500 m spacing is placed inside it, and each grid point is used as a 'sample map point'. These points are not homes, people or population.
3. For each point, the nearest facility in the selected category is found by straight-line distance computed with the Haversine formula (Earth radius 6,371,008.8 m).
4. When a hypothetical facility is added, the same calculation is repeated; it counts as the nearest facility only if strictly closer than the existing nearest one.
5. Categories (swimming facility, park) are never substituted for each other. Facilities outside the study area can be the nearest one if they lie inside the extraction box.
6. The 'canal belt' is the set of sample points 500 m or closer to a mapped canal segment.

10. Limitations and uncertainties

- Distances are not walking distances or times; roads, bridges, the canal and traffic barriers are not considered.
- OpenStreetMap may be incomplete: unmapped facilities are not reflected; operating status, fees and public access of mapped facilities are not verified.
- Sample points do not represent population; the results do not answer 'how many people'.
- This tool does not predict drowning risk or the number of cases prevented.
- Land availability, ownership, cost, permits and technical feasibility of hypothetical facilities are not assessed.
- Closure of any specific canal section is not verified.

11. Sources

- **[VERIFIED]** Adana Büyükşehir Belediye Meclisi, karar sayı 67597785-105.04/183, 12.06.2026. https://www.adana.bel.tr/panel/uploads/mecliskararlari_v/files/2026-haziran-ayi-pdf.pdf
- **[VERIFIED]** Adana Büyükşehir Belediyesi haber sayfası, 21.07.2025. https://www.adana.bel.tr/tr/haber/27-bin-684-cocuga-ucretsiz-spor-egitimi-veriliyor
- **[VERIFIED]** T.C. Tarım ve Orman Bakanlığı — Adana Zirai Üretim İşletmesi Tarımsal Yayım ve Hizmetiçi Eğitim Merkezi, 10.09.2025. https://egitim.tarimorman.gov.tr/adana/Haber/1162/Adana-Tariminda-Yeni-Donem-Basliyor-Acik-Sulama-Kanallari-Tarihe-Karisiyor-Kapali-Basincli-Sulama-Sistemine-Geciliyor
- **[VERIFIED]** Adana Büyükşehir Belediye Meclisi, karar sayı 67597785-105.04/181, 12.06.2026. https://www.adana.bel.tr/panel/uploads/mecliskararlari_v/files/2026-haziran-ayi-pdf.pdf
- **[REQUIRES CONFIRMATION]** Yeni Birlik Gazetesi (yerel basın), 25.07.2026. https://www.gazetebirlik.com/yerel-haberler/adanada-sulama-kanallarinda-denetim-7-kisiye-25-bin-tl-ceza-kesildi/1049295
- OpenStreetMap data: ODbL 1.0 — © OpenStreetMap contributors. https://www.openstreetmap.org/copyright

12. Questions to ask local planners

1. What is the current list and location of public swimming facilities in this area (municipality, provincial directorate, club protocols)?
2. Which of the unnamed pools on the map are public, and which are apartment-complex or hotel pools?
3. Is there a timeline and an urban-use plan for the open canal sections in this corridor during the switch to closed pressurised irrigation?
4. If the canal line is decommissioned, could shaded public space or water-play areas be considered along it? What are the ownership and technical constraints?
5. How far do summer swimming-lesson places reach children in these neighbourhoods?

Sample point details

Point	Nearby neighbourhood (approx.)	To canal	Before	After	Change	Within threshold
A01	Gürselpaşa	1.34 km	1.06 km	1.06 km	no change	No
A02	Gürselpaşa	1.16 km	794 m	794 m	no change	Yes
A03	Pınar	1.11 km	629 m	629 m	no change	Yes
A04	Pınar	1.13 km	305 m	305 m	no change	Yes
A05	Pınar	1.21 km	93 m	93 m	no change	Yes
A06	Tellidere	1.19 km	426 m	426 m	no change	Yes
A07	Tellidere	1.16 km	924 m	924 m	no change	Yes
A08	Yeşilyurt	1.02 km	1.42 km	1.42 km	no change	No
A09	Yeşilyurt	869 m	1.92 km	1.84 km	−80 m	No
A10	Yenibaraj	908 m	2.42 km	2.01 km	−411 m	No
A11	Yenibaraj	575 m	2.92 km	2.28 km	−643 m	No
A12	Kışla	405 m	3.42 km	2.62 km	−806 m	No
B01	Gürselpaşa	871 m	1.46 km	1.46 km	no change	No
B02	Gürselpaşa	684 m	1.07 km	1.07 km	no change	No
B03	Gürselpaşa	606 m	587 m	587 m	no change	Yes
B04	Pınar	649 m	206 m	206 m	no change	Yes
B05	Fatih	717 m	456 m	456 m	no change	Yes
B06	Fatih	696 m	617 m	617 m	no change	Yes
B07	Fatih	673 m	1.03 km	1.03 km	no change	No
B08	Yeşilyurt	628 m	1.49 km	1.30 km	−188 m	No
B09	Yeşilyurt	727 m	1.97 km	1.36 km	−614 m	No
B10	Sümer	717 m	2.46 km	1.58 km	−883 m	No
B11	Sümer	241 m	2.96 km	1.91 km	−1.05 km	No
B12	Ziyapaşa	92 m	3.45 km	2.30 km	−1.15 km	No
C01	Aydınlar	384 m	1.70 km	1.70 km	no change	No
C02	Gürselpaşa	226 m	1.26 km	1.26 km	no change	No
C03	İsmetpaşa	106 m	891 m	891 m	no change	Yes
C04	İsmetpaşa	177 m	701 m	701 m	no change	Yes
C05	Fatih	224 m	831 m	831 m	no change	Yes
C06	Fatih	199 m	1.04 km	1.04 km	no change	No
C07	Mithatpaşa	201 m	1.32 km	1.00 km	−324 m	No
C08	Mithatpaşa	137 m	1.71 km	806 m	−904 m	Yes
C09	Sümer	237 m	2.14 km	894 m	−1.25 km	Yes
C10	Sümer	338 m	2.60 km	1.20 km	−1.40 km	No
C11	Sümer	230 m	3.07 km	1.61 km	−1.46 km	No
C12	Ziyapaşa	245 m	3.55 km	2.06 km	−1.49 km	No
D01	Aydınlar	102 m	1.96 km	1.96 km	no change	No
D02	Barış	208 m	1.60 km	1.60 km	no change	No
D03	Barış	394 m	1.32 km	1.32 km	no change	No
D04	İsmetpaşa	295 m	1.20 km	1.20 km	no change	No
D05	İsmetpaşa	268 m	1.28 km	1.28 km	no change	No
D06	Mithatpaşa	298 m	1.51 km	1.14 km	−369 m	No

Point	Nearby neighbourhood (approx.)	To canal	Before	After	Change	Within threshold
D07	Mithatpaşa	279 m	1.72 km	671 m	−1.05 km	Yes
D08	Namık Kemal	353 m	2.03 km	316 m	−1.71 km	Yes
D09	Namık Kemal	253 m	2.41 km	500 m	−1.91 km	Yes
D10	Gazipaşa	152 m	2.82 km	949 m	−1.87 km	Yes
D11	Gazipaşa	109 m	3.26 km	1.43 km	−1.83 km	No
D12	Ziyapaşa	266 m	3.72 km	1.92 km	−1.79 km	No
E01	Aydınlar	593 m	2.30 km	2.30 km	no change	No
E02	Barış	650 m	2.00 km	2.00 km	no change	No
E03	Barış	890 m	1.79 km	1.79 km	no change	No
E04	Yeşilevler	772 m	1.70 km	1.70 km	no change	No
E05	Bahçelievler	761 m	1.76 km	1.61 km	−144 m	No
E06	Denizli	794 m	1.95 km	1.12 km	−827 m	No
E07	Denizli	774 m	2.16 km	633 m	−1.52 km	Yes
E08	Namık Kemal	843 m	2.41 km	224 m	−2.19 km	Yes
E09	Namık Kemal	742 m	2.74 km	447 m	−2.29 km	Yes
E10	Kurtuluş	642 m	3.11 km	922 m	−2.19 km	Yes
E11	Gazipaşa	608 m	3.51 km	1.41 km	−2.10 km	No
E12	Cemalpaşa	666 m	3.94 km	1.91 km	−2.03 km	No
F01	Fevzipaşa	1.09 km	2.69 km	2.69 km	no change	No
F02	Fevzipaşa	1.13 km	2.44 km	2.44 km	no change	No
F03	Demetevler	1.32 km	2.27 km	2.27 km	no change	No
F04	Demetevler	1.26 km	2.20 km	2.20 km	no change	No
F05	Bahçelievler	1.25 km	2.24 km	1.75 km	−498 m	No
F06	Narlıca	1.29 km	2.40 km	1.30 km	−1.09 km	No
F07	Ahmet Remzi Yüreğir	1.27 km	2.62 km	922 m	−1.69 km	Yes
F08	Ahmet Remzi Yüreğir	1.33 km	2.83 km	707 m	−2.13 km	Yes
F09	Döşeme	1.23 km	3.11 km	806 m	−2.31 km	Yes
F10	Kurtuluş	1.14 km	3.45 km	1.14 km	−2.30 km	No
F11	Kurtuluş	1.11 km	3.81 km	1.57 km	−2.25 km	No
F12	Cemalpaşa	1.14 km	4.21 km	2.03 km	−2.18 km	No